**Course:** Grade 11 Physics (Mechanics) **Date:** 2026-01-22
**Topic:** Systems of Particles **Doc Type:** Solution Key

# Teacher's Solution Key: Assessment Problem 3

## Solution & Marking Scheme (Total: 5 pts)

### 1. Total Mass (1 pt)

- **Answer:** $20$ kg **[1.0 pt]**

$$M_{sys} = 4 + 6 + 5 + 2 + 3 = \mathbf{20} \text{ kg}$$

### 2. Center of Mass Coordinates (3 pts)

- **Abscissa $x_G$ (1.5 pts):**

$$\sum m_i x_i = 4(3) + 6(-2) + 5(-4) + 2(5) + 3(0) \quad \textit{(Correct moments)}$$
$$= 12 - 12 - 20 + 10 = -10 \quad \textbf{[0.5 pt]}$$
$$x_G = \frac{-10}{20} = -\mathbf{0.5} \text{ cm} \quad \textit{(Final Result)} \ \textbf{[1.0 pt]}$$

- **Ordinate $y_G$ (1.5 pts):**

$$\sum m_i y_i = 4(3) + 6(2) + 5(-4) + 2(-2) + 3(5) \quad \textit{(Correct moments)}$$
$$= 12 + 12 - 20 - 4 + 15 = 15 \quad \textbf{[0.5 pt]}$$
$$y_G = \frac{15}{20} = \mathbf{0.75} \text{ cm} \quad \textit{(Final Result)} \ \textbf{[1.0 pt]}$$

### 3. Analysis (1 pt)

- **Answer:** Quadrant II **[1.0 pt]**

Result: $\boxed{G(-0.5, 0.75)}$. Since $x < 0$ and $y > 0$, it is in **Quadrant II**.